

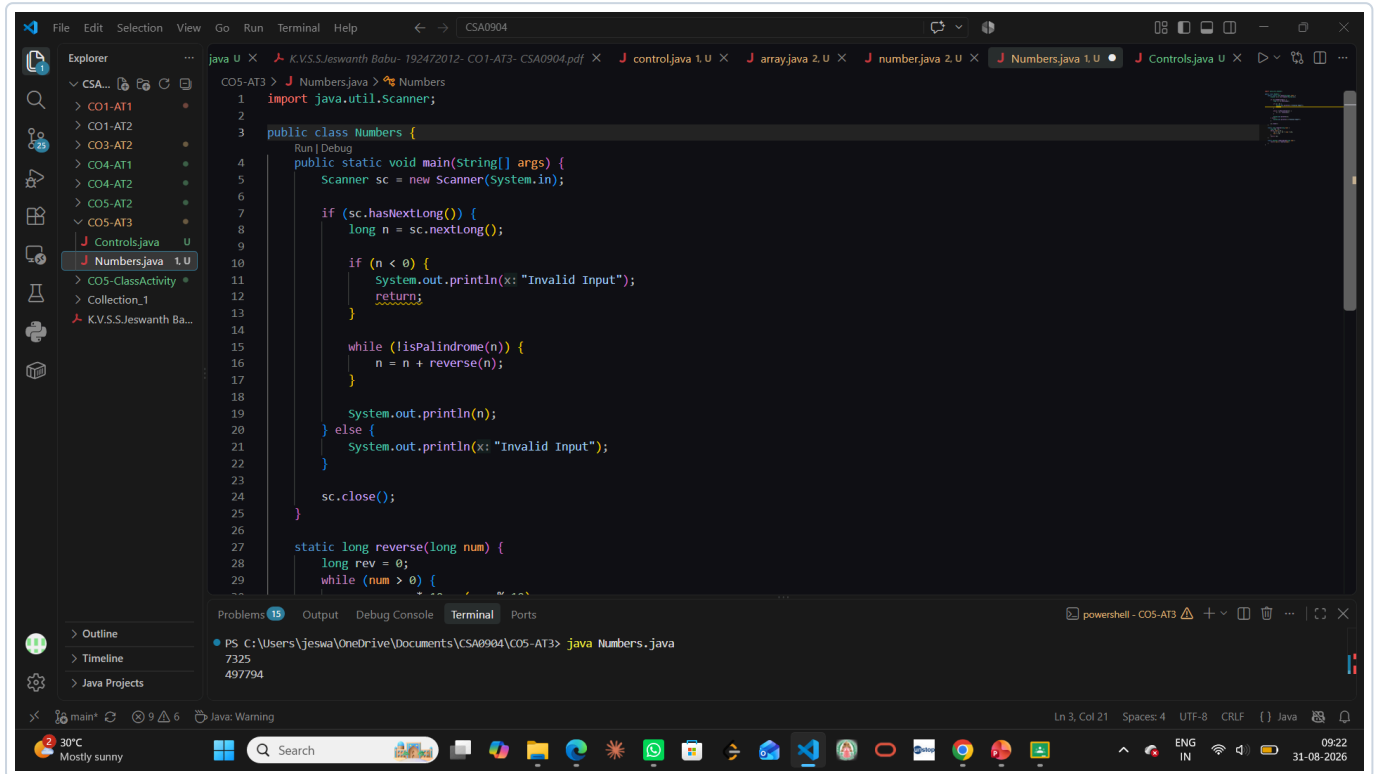
CO5-AT3: LEVEL EXAM-3 ASSESSMENT

Java Programming Laboratory Documentation

Course Code	CSA0904
Course Name	Java Programming
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Assessment Topic	Reverse and Add to Obtain Palindrome & Roman Numeral Conversion
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QUESTION 1: REVERSE AND ADD TO FORM PALINDROME (NUMBERS.JAVA)

Problem Statement: Write a Java program that accepts a number, reverses it, and adds the reversed number to the original number repeatedly until the resultant sum becomes a palindrome. Handle non-negative constraints and invalid input formats gracefully.



The screenshot shows a Java IDE with the following code in `Numbers.java`:

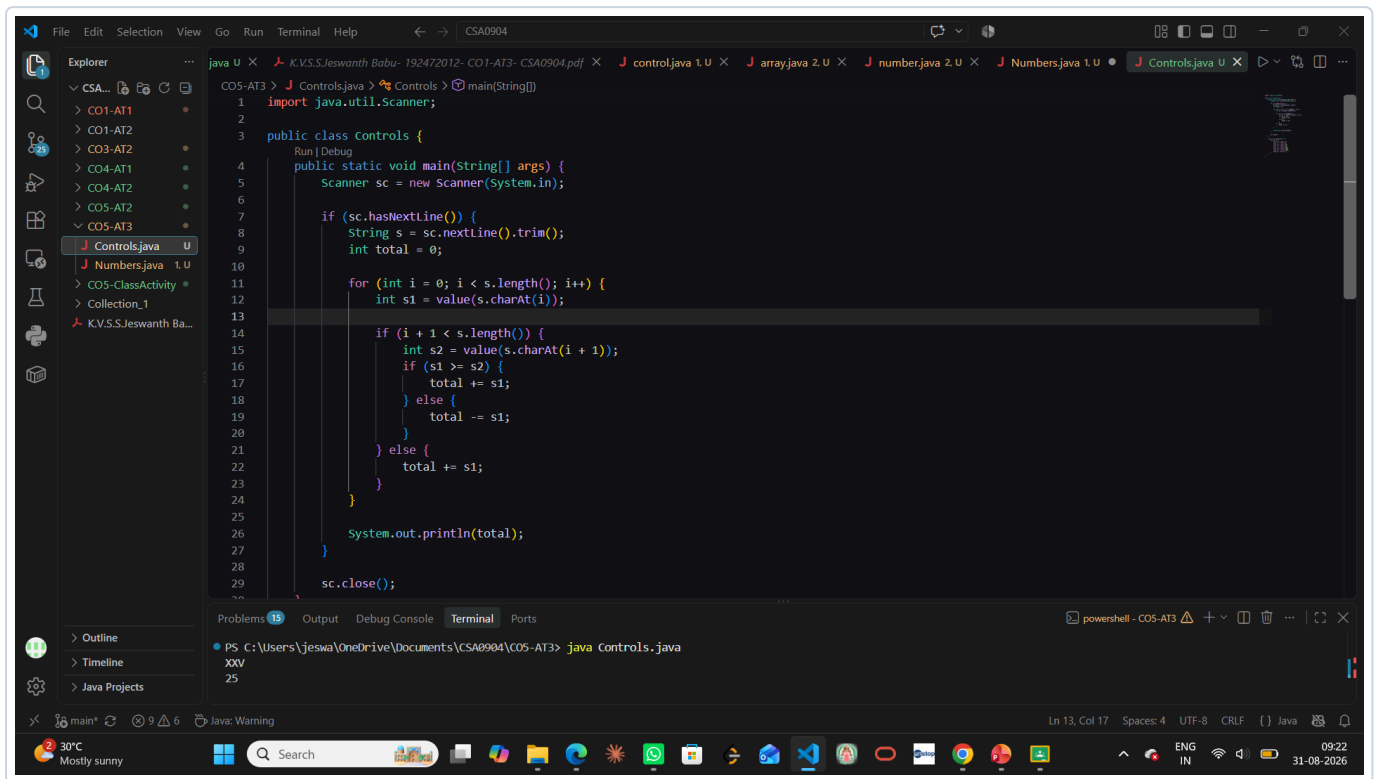
```
1 import java.util.Scanner;
2
3 public class Numbers {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         if (sc.hasNextLong()) {
8             long n = sc.nextLong();
9
10            if (n < 0) {
11                System.out.println(x: "Invalid Input");
12                return;
13            }
14
15            while (!isPalindrome(n)) {
16                n = n + reverse(n);
17            }
18
19            System.out.println(n);
20        } else {
21            System.out.println(x: "Invalid Input");
22        }
23
24        sc.close();
25    }
26
27    static long reverse(long num) {
28        long rev = 0;
29        while (num > 0) {
30            rev = rev * 10 + num % 10;
31            num = num / 10;
32        }
33        return rev;
34    }
35
36    static boolean isPalindrome(long num) {
37        long rev = reverse(num);
38        return num == rev;
39    }
40}
```

The terminal output shows the execution of the program:

```
PS C:\Users\jeswa\OneDrive\Documents\CSA0904\CO5-AT3> java Numbers.java
7325
497794
```

QUESTION 2: ROMAN NUMERAL TO INTEGER CONVERSION (CONTROLS.JAVA)

Problem Statement: Write a Java program to convert a Roman numeral string into its equivalent decimal integer value by processing character values and applying subtractive/additive notation rules.



The screenshot shows an IDE with a Java file named `Controls.java` open. The code implements a method to convert a Roman numeral string to an integer. The logic involves iterating through the string and checking if the current character is less than the next character. If so, it subtracts the current character's value from the total; otherwise, it adds it. The values for Roman numerals are stored in a static array.

```
1 import java.util.Scanner;
2
3 public class Controls {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         if (sc.hasNextLine()) {
8             String s = sc.nextLine().trim();
9             int total = 0;
10
11             for (int i = 0; i < s.length(); i++) {
12                 int s1 = value(s.charAt(i));
13
14                 if (i + 1 < s.length()) {
15                     int s2 = value(s.charAt(i + 1));
16                     if (s1 >= s2) {
17                         total += s1;
18                     } else {
19                         total -= s1;
20                     }
21                 } else {
22                     total += s1;
23                 }
24             }
25             System.out.println(total);
26         }
27         sc.close();
28     }
29 }
```

The terminal output shows the execution of the program with the input `XXV` resulting in the output `25`.

```
PS C:\Users\jeswa\OneDrive\Documents\CSA0904\CO5-AT3> java Controls.java
XXV
25
```